Test 1 : Sequences and Series

EXERCISE 1 The speed of light is 300 000 kilometres per second. The average distance from the Sun to the Earth is 149.6 million km.

A light-year is the distance light travels in one year and is equal to 9 467 280 million km.

Polaris is a bright star, visible from the Northern Hemisphere. The distance from the Earth to Polaris is 323 light-years.

1. Calculate the time, in minutes, it takes for light from the Sun to reach the Earth.
2. Find the distance from the Earth to Polaris in millions of km. Give your answer in the form $a \times 10^k$ with $1 \leq a < 10$ and $k \in \mathbb{Z}$.

EXERCISE 2 Find, in simplest form, a formula for the general term u_n of :

(a) 86, 83, 80, 77, ...

(b) $\frac{3}{4}$, 1, $\frac{7}{6}$, $\frac{9}{7}$, ...

EXERCISE 3 Insert four numbers between 3 and 12 so that all six numbers are in arithmetic sequence.

EXERCISE 4 A farmer feeds his cattle herd with hay every day in July. The amount of hay in his barn at the end of day n is given by the arithmetic sequence

$$u_n = 100 - 2.7n \quad \text{tonnes.}$$

1. Write down the amount of hay in the barn on the first three days of July.
2. Find and interpret the common difference.
3. Find and interpret u_{25}.
4. How much hay is in the barn at the beginning of August ?

EXERCISE 5 The initial population of rabbits on a farm was 50. The population increased by 7% each week.

1. How many rabbits were present after :
   a. 15 weeks
   b. 30 weeks

2. How long would it take for the population to reach 500 ?

EXERCISE 6 How much should I invest now if I need a maturing value of 10 000 in 4 years' time, and I am able to invest at 8% p.a. compounded twice annually ? Give your answer to the nearest cent.

EXERCISE 7 ONLY SL

1. Find the sum of $\frac{1}{2} + 3 + \frac{5}{2} + 8 + \cdots$ to 50 terms.

2. Evaluate

$$\sum_{k=0}^{13} (2 - 0.3k).$$

EXERCISE 8 ONLY HL Helen and Jane both begin new jobs each starting on an annual salary of $70,000. At the start of each new year, Helen receives an annual salary increase of $2400.

Let H_n represent Helen's annual salary at the start of her nth year of employment.

At the start of each new year, Jane receives an annual salary increase of 3% of her previous year's annual salary.

Jane's annual salary, J_n, at the start of her nth year of employment is given by

$$J_n = 70\,000(1.03)^{n-1}.$$

At the start of year N, Jane's annual salary exceeds Helen's annual salary for the first time.

1. Show that

$$H_n = 2400n + 67\,600.$$

2. Given that J_n follows a geometric sequence, state the value of the common ratio, r.

3. **a.** Find the value of N.

b. For the value of N found in part (3)(a), state Helen's annual salary and Jane's annual salary, correct to the nearest dollar.

4. BONUS : Find Jane's total earnings at the start of her 10th year of employment. Give your answer correct to the nearest dollar.